Basics of automation technology

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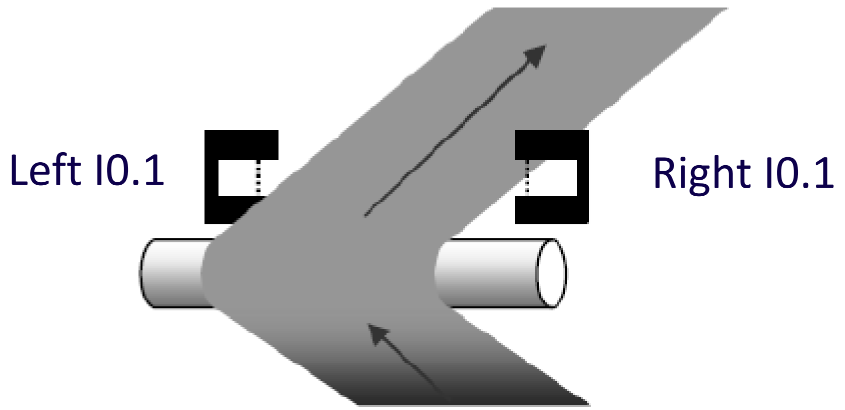
Example 1

Assignment: Make a program that can be used to indicate when the packaging plastic moves too far from the middle (left or right line). Left and right of the line has a light gate (I0.1 and I0.2) and an indicator light is used for the indication(Q0.2). The alarm is given only when the device is running (running information I1.0.



Extra: The alarm should be set up but it can be reset using the button connected to input I0.3 if the error situation is not active.

Example 2

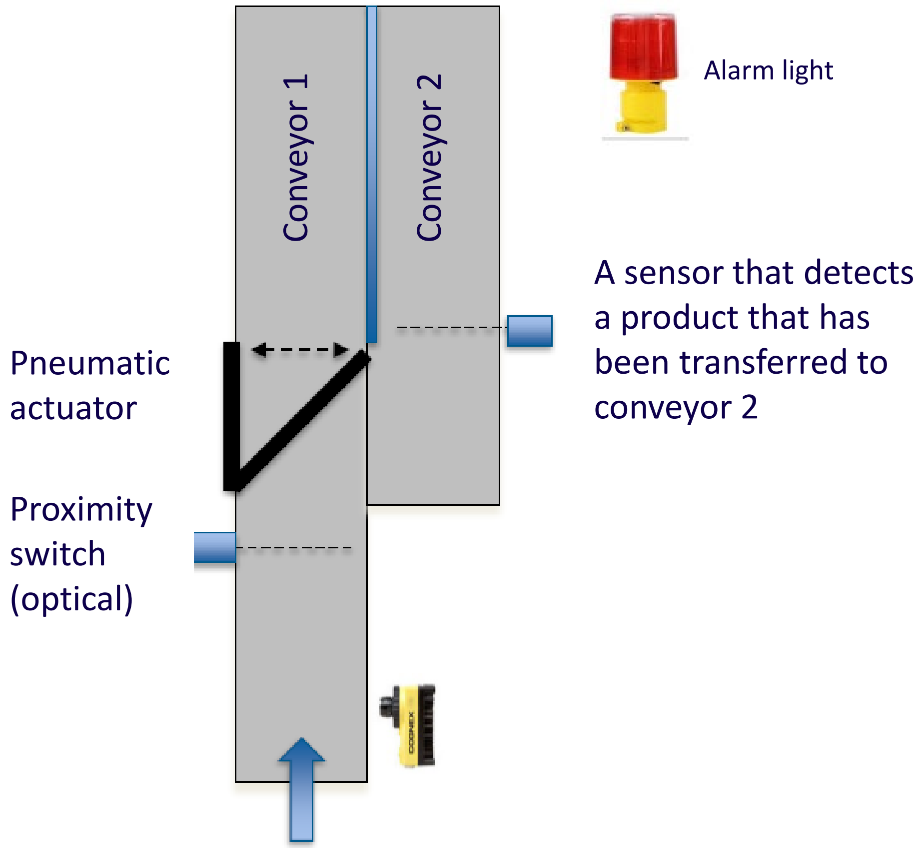
Conveyor control and removal of defective products to the conveyor 2

The operation of the device is described in the next slides:

Conveyor 1

Conveyor 2

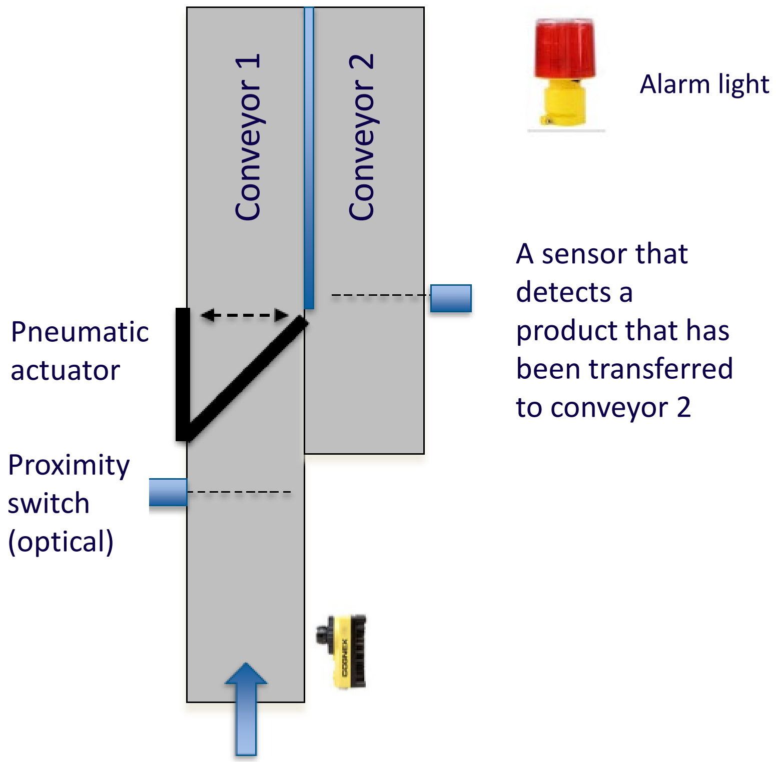
Shorting



Conveyor 1

Conveyor control, "Conveyor 1":

- Start button (normally open contact) is connected to input I0.0
- Start button (normally open contact) is connected to input I0.1
- Enable button is connected to input I0.2
- Conveyor 1 is controlled with output Q0.0
- Conveyor 1 starts if Enable is True and Start button is pressed
- Conveyor stops if Stop button is pressed or Enable is missing



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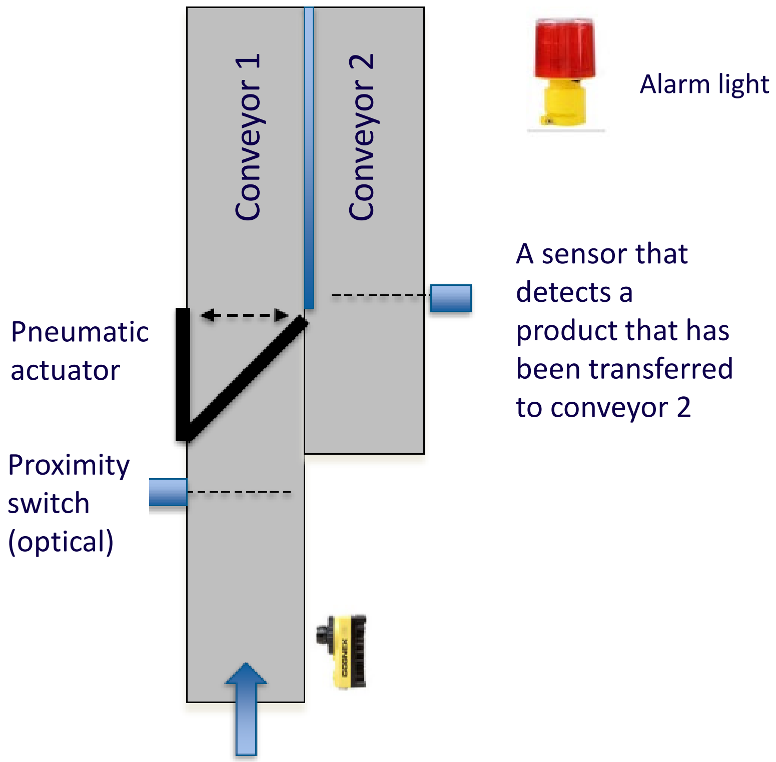
Conveyor 2

Conveyor control, "Conveyor 2":

Conveyor 2 starts if:

- Conveyor 1 is on
- Conveyor 2 stops on the same terms as conveyor 1 but with 10 second off delay.
- Conveyor 2 is controlled with output Q0.1



Shorting

Sorting non-currant products for conveyor 2: The pneumatic cylinder turns the missile, which directs defective products to conveyor 2.

I

- An optical sensor connected to input I0.3 detects that a product is approaching the missile
- Input I0.4 is connected to the running data of conveyor 1
- Input I0.5 is connected to the running data of conveyor 2
- An optical sensor is connected to input I0.6, which recognizes that the product has been successfully directed to conveyor 2.
- A pressure switch is connected to the input I0.7, which monitors the pressure in the compressed air network supplying the cylinder. I0,7 = 1, provided that the pressure is at a sufficient level. If the pressure disappears, the conveyors must stop and a red error light (Q0.3) must also be switched on
- A signal from the quality control equipment is connected to input I1.0. If the product is defective I1.0 = 0.

Q

- A directional valve controlling the pneumatic cylinder is connected to output Q0.2.
- Red alarm light is connected to output Q0.3

- The missile must turn if both conveyors are running and the defective product is approaching the missile. The missile is allowed to return to the initial position after the defective product has successfully moved to conveyor 2.